

From rough building hull to evidence-based renovation workflow

This is not a finished BIM model or complete digital twin. It is a local-first planning workflow that organizes evidence, uncertainty and decisions around an existing building.

LOCAL-FIRST

BIM/IFC/AAS-INSPIRED

SITE VERIFICATION REQUIRED

EXPERT VALIDATION REQUIRED

MAPPED FROM PLANNING SEED

- Early geospatial context and rough hull geometry are now treated as a structured baseline, not as measured reality.
- Facts, assumptions, measurements, risks and decisions are split into registers so uncertainty stays visible.
- The agentic layer prepares better questions for architects, engineers, energy consultants and craftspeople.

1

Geodata and rough hull experiment

Generated model outputs, terrain context and IFC-oriented export experiments establish a first spatial reference.

2

Evidence registers

Building facts, assumptions, missing measurements, findings, risks and decisions become local project data.

3

Uncertainty-aware reasoning

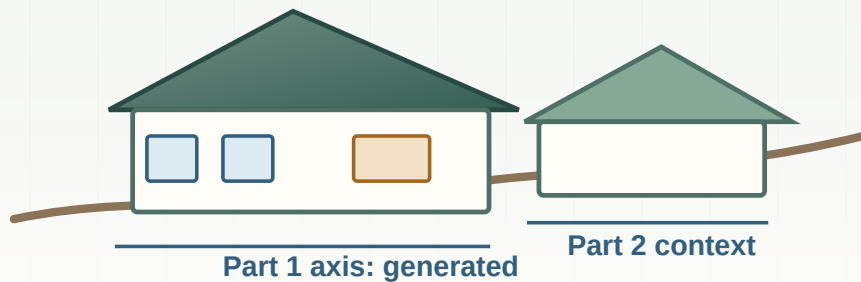
Rule-based summaries separate facts from assumptions and produce practical next actions for site visits.

Geometry and model outputs are useful when their uncertainty is explicit

Sanitized building context view

Recreated from current app concepts. No address, coordinates, land-record identifiers, private photos or exact property identifiers are included.

Estimated terrain line
Generated hull baseline



GENERATED MODEL OUTPUT

NEEDS SITE CONTROL

NOT CONSTRUCTION-READY

Confirmed hull parts

confirmed building part
selection

Part 1 and Part 2 parts

DOCUMENTED

Part 1 main body length

Part 1 schematic floor plan / LoD2-derived local axis

16.88 m

GENERATED

Nominal storey height seed

vertical CAD seed assumption

2.4 m

ESTIMATED

Keller schematic size

owner-provided schematic
interpretation captured in Part
1 plan

approximately 4 x 7 m

ESTIMATED

Metadata validation foundation

metadata-first
sprint

schemas, examples,
references, privacy and
modeling checks package

DOCUMENTED

Facts, assumptions and missing measurements must not be mixed together

Known facts

DOCUMENTED

Confirmed hull parts

Part 1 and Part 2 parts. Source: confirmed building part selection.

GENERATED

Part 1 main body length

16.88 m. Source: Part 1 schematic floor plan / LoD2-derived local axis.

GENERATED

Part 1 main body width

8.08 m. Source: Part 1 schematic floor plan / LoD2-derived local axis.

Open assumptions

NEEDS SITE CHECK

Part 1 exterior wall thickness is currently estimated at about 0.5 m

Wall thickness affects U-value reasoning, opening details, material quantities and CAD seed accuracy.

OPEN

Existing roof build-up, ventilation and underlay/sarking are unknown

Roof insulation and PV decisions can fail if moisture behavior and structural build-up are misunderstood.

NEEDS SITE CHECK

Side-elevation terrain lines are simplified reasoning aids, not measured site profiles

Drainage, plinth moisture and threshold decisions depend on real local ground levels.

Needed site checks

CRITICAL

Identify roof covering, underlay/boarding, ventilation path and visible timber structure

Use S11-S14 photo blocks, measure one rafter bay and document visible layers.

HIGH

Measure representative exterior wall thickness at a window or door reveal

Use tape/laser at an open reveal and photograph the measurement.

HIGH

Verify eaves, ridge, threshold and terrain relationships

Laser-distance photo blocks plus one reliable vertical or diagonal control where safe.

Renovation choices become clearer when dependencies and risks are visible

BLOCKED**Choose roof insulation strategy****DEPENDS ON**

Measure representative exterior wall thickness at a window or door reveal. Identify roof covering, underlay/boardings, ventilation path and visible timber structure.

EXPERT CHECK

Inspect roof build-up, document ventilation paths and validate with an energy consultant.

KEY UNCERTAINTY

A roof insulation build-up could trap moisture if ventilation and existing layers are misunderstood.

NEXT STEP

Inspect roof build-up and timber condition before narrowing insulation options.

UNDER REVIEW**Define realistic PV placement and mounting constraints****DEPENDS ON**

Identify roof covering, underlay/boardings, ventilation path and visible timber structure. Verify eaves, ridge, threshold and terrain relationships.

EXPERT CHECK

Verify roof surfaces, setbacks, windows, penetrations and structure before layout decisions.

KEY UNCERTAINTY

PV suitability could be overestimated if roof condition, obstructions and maintenance zones are not verified.

NEXT STEP

Verify roof condition, obstructions, setbacks and structural plausibility on site.

BLOCKED**Prioritize moisture and drainage interventions****DEPENDS ON**

Document all gutter/downpipe discharge points and ground fall at plinth zones. Confirm Keller size, vault direction, entrance and moisture pattern.

EXPERT CHECK

Document drainage discharge, splash zones, Keller moisture and terrain fall.

KEY UNCERTAINTY

Plinth or Keller moisture could be misclassified without drainage and rainfall evidence.

NEXT STEP

Collect drainage, plinth and Keller evidence during first site visit.

UNDER REVIEW**Frame roof extension, dormer and overhang concepts for expert review****DEPENDS ON**

Verify eaves, ridge, threshold and terrain relationships.

EXPERT CHECK

Treat geometry concepts as discussion aids and require structural expert validation.

KEY UNCERTAINTY

Generated roof and dormer/overhang concepts do not prove structural feasibility.

NEXT STEP

Use side-elevation measurements to decide whether the concept is worth expert review.

Useful AI assistance stays grounded in local evidence and visible uncertainty

Rule-grounded reasoning output

Facts:

Confirmed hull parts: Part 1 and Part 2 parts (documented, source: confirmed building part selection)

Assumptions:

Part 1 exterior wall thickness is currently estimated at about 0.5 m. Verification: Measure a representative reveal at a window or exterior door.

Risks:

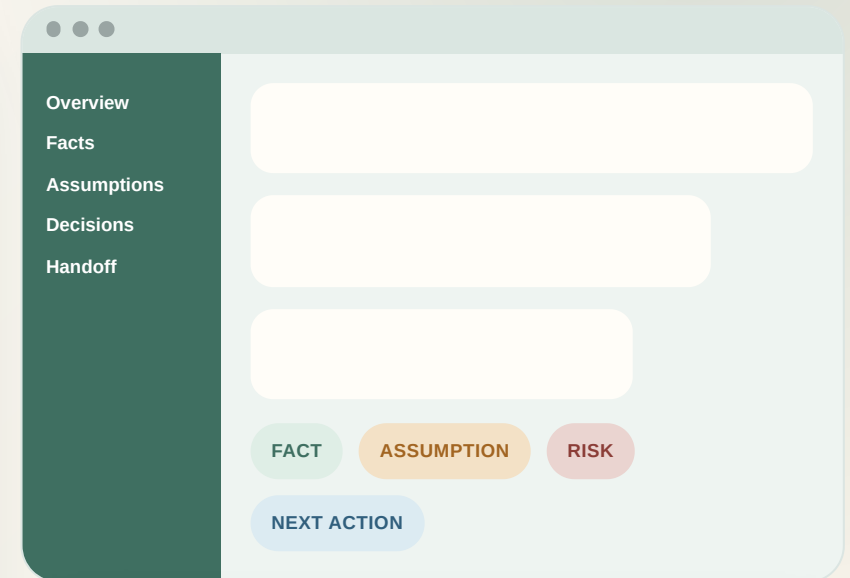
A roof insulation build-up could trap moisture if ventilation and existing layers are misunderstood.
Mitigation: Inspect roof build-up, document ventilation paths and validate with an energy consultant.

Missing information:

Identify roof covering, underlay/boardings, ventilation path and visible timber structure. Method: Use S11-S14 photo blocks, measure one rafter bay and document visible layers.

Suggested next actions:

Inspect roof build-up and timber condition before narrowing insulation options.



NEXT USEFUL SITE VISIT

S01-S20 photo blocks with multiple photos per station and laser distance to a clearly described reference point.

NEXT EXPERT CONVERSATION

Roof build-up, moisture strategy, structural plausibility and PV-ready renovation sequencing.

NEXT IMPLEMENTATION TASK

Import real site-visit evidence into local-only registers and diff it against the previous generated state.